

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

2nd Semester of B. Pharm. Examination

University Theory Examination October 2016

PH109 Pharmaceutical Engineering-II

Date: 22/10/2016, Saturday Time: 10:00 a.m to 01:00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

- Q 1** Attempt any TWO of the following
- A Classify the various filtration devices. Explain the construction, working and advantages of filter press. 10
- B Briefly describe the principle of centrifugation along with various centrifuges used in Pharmacy. Give the factors affecting the same. 10
- C Give the basic concept of phase equilibrium and describe the various factors affecting evaporation. 10
- Q 2** Attempt any TWO of the following
- A Describe the various factors affecting the rate of filtration and explain the mechanism of filtration process. 10
- B Classify the evaporators. Explain single effect and multiple effect evaporators in detail. 10
- C(I) Give the construction, working and advantages of sedimentation type Centrifuge. 5
- (II) Explain Duhring's Rule and Raoult's law. 5

SECTION- II

- Q 3** Attempt any TWO of the following
- A Explain the principle, construction, working and advantages of fluidized bed dryer with neat diagram. 10
- B What is flash distillation? Discuss the principle, construction, working, advantages, disadvantages and uses of flash distillation. 10
- C Explain the principle and working of air conditioner with neat diagram. 10
- Q 4** Attempt any TWO of the following
- A Explain the construction, working, advantages and disadvantages of spray dryer. 10
- B Write the applications, theory and principle of refrigeration cycle. 10
- C What is molecular distillation? Explain the principle, construction and working of centrifugal molecular still. 10
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